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Key drivers in the behavior of potential consumers of remanufactured products: a study on laptops in Spain

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ABSTRACT

Many companies have begun to implement end-of-use strategies (remanufacturing, recycling, and reuse) in response to worldwide concern about environmental issues. Simultaneously, academic research on remanufacturing has turned to focus on the analysis of activities from a supply perspective, *i.e.*, collection of end-of-use products, reverse logistics, disassembly, assembly, etc. Very little attention has been paid, however, to the demand side of remanufactured products, in particular, to the consumer profile of these products or to determining their most effective marketing strategies. Companies interested and involved in fostering the demand for remanufactured products need to be aware of the importance of knowledge of their potential consumers in order to most suitably manage their remanufacturing and marketing activities. Given this context, the principal objective of the present work was to make a first approach to determining basics characteristics of the profile of potential consumers of remanufactured products. To this end, an empirical analysis was conducted of the key variables underlying the purchase intentions of potential consumers in order to gain insight into the key drivers of that behavior. The instrument used was a questionnaire elaborated for application to potential consumers of a specific product (laptops) in two Spanish universities. The results revealed the drivers that explain the self-declared purchase intention of a remanufactured laptop, and provide managers of firms interested in implementing green initiatives in their supply chain with useful information for their consideration of closed-loop supply chains and for integrating marketing decisions concerning remanufactured products into the development of end-of-use strategies.

1. Introduction

The relationship of environmental issues with industrial and business activities has been a major topic of research interest since the 1990s (Seuring et al., 2008). During this time, various approaches having been taken to studying environmental problems in the context of the supply chain (Carter and Rogers, 2008; Hassini et al., 2012; Seuring, 2013; Seuring and Müller, 2008; Srivastava, 2007). Of particular interest is the so-called Closed-Loop Supply Chain (CLSC) (Guide and Van Wassenhove, 2009; Ilgin and Gupta, 2010; Pokharel and Mutha, 2009; Rubio et al., 2008). There is a broad consensus that one of the challenges for CLSC research in the coming years is the need to examine in depth its relationships with the market and consumers. In this specific area of research, most of the problems analyzed in the literature have been approached from the supply side of end-of-life (EOL) and end-of-use (EOU) products. There have been studies of the flow of goods from the consumer back to the producer or to the recovery agent, *e.g.*, EOU collection, return, value recovery, 3-R (reduce, reuse, recycle), inventory management, etc. There has been little work, however, on such issues as the marketing of recovered products, their acceptance by consumers, the existence of new markets for these products and how firms can promote these markets, what marketing strategies are best suited for this purpose, or what type of consumer should be targeted. The present work is an attempt to provide some input in this regard through an analysis of the behavior of the consumer of a remanufactured product. To this end, we conducted a survey of potential consumers of a remanufactured product (laptops), the aim being to identify the key drivers that "define" a profile for this type of consumer, from which firms can design and implement the most appropriate marketing strategies.

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1.1. Literature review

Several studies in marketing and consumer behavior have focused on the analysis of topics related with used products and their relationship with their original counterparts, such as the connection between markets for new products and markets for used products (Purohit, 1992; Thomas, 2003), the buying and selling behavior of consumers of used products (Hickey and Fitzpatrick, 2008; Lee and Lee, 2005), the factors which may influence the price policy (Brough and Isaac, 2012; Quariguasi-Frota-Neto et al., 2012), or the distribution strategy (Purohit and Staeling, 1994) for these type of products, among others. In this case, we are interested in the marketing issues of remanufacturing.

Definitions of remanufacturing do not usually take into account aspects related to the subsequent marketing of the remanufactured product. Instead, they only tend to highlight the technical issues of the processes needed for remanufacturing, and the fact that the remanufactured product must have a condition similar to that of the original product. One should not forget, however, that at the end of the remanufacturing process the product has to be made available to consumers and result in them purchasing it, *i.e.*, remanufacturing is a process whose ultimate goal is the product's sale so that its value can be recovered. Otherwise, what would be the point of remanufacturing? Indeed, in their definition of the concept, Atasu et al. (2010) state that: "*Remanufacturing operations involve taking used products, bringing them back to as-new condition, and selling them again, often with exactly the same warranty as a new product.*"

Undoubtedly, as noted above, while remanufacturing has generated much attention among academics, researchers, and industry and commerce, aspects related to the marketing phase of remanufactured products have generally remained outside the focus of attention of the major works published on this topic. In an interesting study on the evolution and challenges of CLSC, Guide and Van Wassenhove (2009) argue very precisely that one of the challenges of this concept for the coming years is to develop interdisciplinary research connecting with other areas of knowledge such as marketing and accounting. They stress that: "*If prices and markets are not fully understood, they become barriers, no matter how well the operational system is designed. This research phase has barely begun to investigate these issues.*" Subramoniam et al. (2009) also emphasize this in stating that: "*There is often a lack of proper technical, environmental, and quality data within original equipment divisions to effectively convince new customers to use remanufactured products.*" Atasu et al. (2010) analyze the effects of remanufacturing in cannibalizing the original products, and identify the gap in research in asserting that "*the marketing aspects of remanufacturing are largely ignored by academic research*", and the importance of the commercial aspects of remanufacturing in noting that "*optimal use of remanufacturing requires understanding of how customer values remanufactured product.*" Ilgin and Gupta (2012) re-emphasized this need for research in stating that: "*Remanufactured products are usually not included in the strategic selling plan. There is a need to educate consumers that remanufactured products can be as good as brand new products.*"

With respect to the marketing issues of remanufacturing, most research – see Agrawal et al. (2012) or Subramanian and Subramanyan (2012) for a more detailed review of the literature – has been related to such topics as willingness to pay for remanufactured products (*e.g.*, Essoussi and Linton, 2010; Michaud and Llerena, 2011; Occhinnikov, 2011), effects of cannibalization (Atasu et al., 2010; Guide and Li, 2010; Guide and Van Wassenhove, 2001), competition (Atasu et al., 2008; Debo et al., 2005; Majumder and Groenevelt, 2001), consumer perception of remanufactured products (Hazen et al., 2012), and pricing decisions (Subramanian and Subramanyan, 2012).

This paper tries to make a contribution to the literature in this field by analyzing issues that have been traditionally unexplored, and for which academics and practitioners are claiming for more research, particularly in issues as consumer behavior and the market for remanufactured products (Souza, 2013; Atasu et al., 2010; Guide and Van Wassenhove, 2009).

1.2. Objectives and structure

With the present communication, our goal is to contribute to the development of research in the CLSC field from a marketing perspective – specifically, from the point of view of consumer behavior. Given this context, the principal objective of the present work was to make a first approach to determining basic characteristics of the profile of potential consumers of remanufactured products, so that interested firms may have a solid base of information on which to establish their policies for the development and strengthening of these markets. By remanufactured products we understand those which, once they cease to meet the needs of consumers either because they do not work or because they have become obsolete, can be recovered and undergo certain processes that give them back the same functionality as their original equivalents, and are then offered for sale with characteristics of quality, performance, warranty, and after-sales services similar to those of the original products, and generally at a lower price.

Thereby, one will not only be endowing the process of recovery of EOU products with economic meaning (financial profit, remanufacturing to sell), but also actually *closing* the CLSC cycle by putting recovered products back on the market. In a sense, one would also be contributing to strengthening the academic links between marketing and operations management by stimulating new joint lines of inquiry.

The rest of the paper is organized as follows. Section 2 describes the methods, the model, and the hypotheses of the study, together with the data obtained from the survey. Then, in Section 3, we present the results of a structural model, followed in Section 4 with a discussion of those results. Finally, Section 5 sets out the main conclusions drawn from the results of the study.

2. Methods

2.1. The model

The proposed model is based on the Theory of Planned Behavior (Ajzen, 1985; Ajzen, 1991; Fishbein and Ajzen, 1975), which has been used successfully in other studies to explain consumer behavior in situations similar to that considered in the present work. An example is the behavior of consumers of organic products (Aragón and Llorens, 1997; Arvola et al., 2008; Chan, 2001; Gill et al., 1986; Kaisser et al., 1999; Kalafatis et al., 1999; Smith and Paladino, 2010; Vermeir and Verbeke, 2007). Our aim here will be to formulate a model capable of explaining what kind of consumer would be willing to purchase a particular remanufactured product, specifically, a laptop computer. This should then be of assistance to firms in the sector by providing them with a set of guidelines for the establishment of effective marketing strategies for this type of product. The goal with the proposed model (Fig. 1) is to describe the *Intention to Purchase a Remanufactured Laptop* (Purchase Intention, PI) on the basis of four variables that might influence that intention:

1. The *Attitude Towards Purchasing a Remanufactured Laptop* (AP) refers to the degree to which a person has a favorable or unfavorable evaluation or appraisal of the purchasing of a remanufactured laptop. As a general rule, the more favorable the

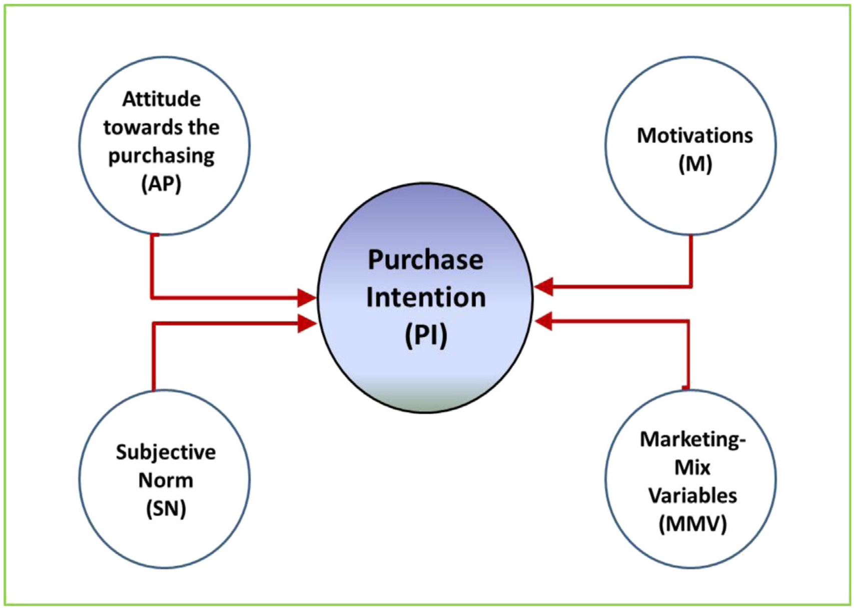

Fig. 1. The model.

attitude the stronger should be the intention to perform this behavior.

2. The variable *Subjective Norm* (SN) is related to the social pressure perceived by the individual as to whether or not to perform a certain behavior. In our case, it reflects the individual's perception of what their closest referents (family, friends, classmates, workmates, etc.) think they should do about whether to purchase a remanufactured laptop.
3. The variable *Motivations* (M) indicates the reasons that may lead an individual to carry out this particular type of behavior (purchase a remanufactured laptop) at some time in the future, such as the fact that the remanufactured laptop is marketed at a lower price than the original.
4. The so-called *Marketing Mix Variables* (MMV), *i.e.*, price, product, promotion, and distribution. These are included in the model because we believe that they might influence the consumer's intention to purchase. They also constitute an effective tool for firms in their effort to achieve more efficient marketing of these products.

With these premises, our aim with the proposed model of Fig. 1 is to define the profile of consumers of this type of remanufactured product on the basis of the results of testing the four hypotheses to be formulated in the following subsection.

2.2. Hypotheses

As noted above, the variable PI which is the object of the study refers to the greater or lesser likelihood that a consumer will purchase this product rather than the original one, being aware of the main characteristics of remanufactured products. For each of the model's variables, we constructed a measurement scale based on an exhaustive review of the remanufacturing literature and on recommendations for the elaboration of a Theory of Planned Behavior type questionnaire (Ajzen, 2006).

Thus, the variable PI (*Intention to Purchase a Remanufactured Laptop*) was measured by the following 2-item scale. Item PI1 was based on an experiment that simulated the purchasing act of a consumer interested in acquiring a laptop. Three scenarios were presented, consisting of the offer of identical products in terms of performance and technological characteristics, but different in their origin of manufacture (original, remanufactured by the Original Equipment Manufacturer, OEM, and remanufactured by an independent professional) and in their price. The responses were used to estimate for each respondent the probability that they would purchase a remanufactured laptop. Item PI2 was based on the result of asking the respondents to indicate their likelihood of purchasing a remanufactured laptop considering all the basic information (price, quality, performance, etc.) they had about the product as well as the information they had picked up from reading the entire questionnaire.

The Theory of Planned Behavior argues that the more favorable an individual's attitude is to a particular behavior the stronger will be the intention to perform that behavior. Thus, we consider the following hypothesis concerning the variable AP (*Attitude towards Purchasing a Remanufactured Laptop*):

H1. A favorable attitude towards purchasing a remanufactured laptop will positively influence the purchase intention for that class of remanufactured product.

To measure this variable, the respondents were asked to score on a 5-point Likert scale their predisposition to purchasing this type of product.

The variable *Subjective Norm* (SN) was measured by a scale in which the respondents were asked to show their level of agreement or disagreement with two items relating to the views and beliefs of the respondent's closest referents. The corresponding hypothesis was:

H2. A favorable Subjective Norm towards purchasing a remanufactured laptop will positively influence the purchase intention for that class of remanufactured product.

The variable *Motivations* (M) was evaluated by a scale based on a review of the literature identifying the main reasons that might influence the consumption of remanufactured products: technology (Ferrer and Whybark, 2003; Hauser and Lund, 2003; Pickett-Baker and Ozaki, 2008), price (Agrawal et al., 2012; Atasu et al., 2010; Guide and Li, 2010; Mukherjee and Mondal, 2009;

Subramanian and Subramanyam, 2012), and environment (Atasu et al., 2010; Essoussi and Linton, 2010; Janse et al., 2010). The corresponding hypothesis was:

H3. The level of Motivations towards purchasing a remanufactured laptop will positively influence the purchase intention for that class of remanufactured product.

For the *Marketing Mix Variables* (MMV), the respondents were asked to indicate what level of importance the four indicators used in this scale (price, product, promotion, and distribution) had for them. The construction was based on a literature review to identify the relationships between remanufactured product consumption and these marketing mix variables. Various authors (Agrawal et al. 2012; Essoussi and Linton, 2010; Hauser and Lund, 2003) argue that characteristics associated with the product (design, brand, quality, performance, etc.) and its price (Agrawal et al., 2012; Mukherjee and Mondal, 2009; Subramanian and Subramanyan, 2012), the distribution channels used for its marketing (Janse et al., 2010; Subramanian and Subramanyan, 2012), and the various communication tools used to promote it (Majumder and Groenevelt, 2001; Subramanian & Subramanyan, 2012) are taken into account by consumers in purchasing situations concerning remanufactured products. The corresponding hypothesis was:

H4. There is a relationship between the Marketing Mix Variables for remanufactured laptop marketing and the purchase intention for these remanufactured products.

2.3. *The sample and the questionnaire*

According to the exploratory character of this research, a convenience sample comprised 1529 undergraduates of two Spanish universities has been considered. The participants responded to a self-administered questionnaire during April and May 2011. Prior to application of the definitive questionnaire, two pre-tests were conducted. In the first, the questionnaire was evaluated by a group of experts in the disciplines of "Green Marketing" and "Consumer Behavior" to ensure that all the variables and their elements of analysis were correctly represented in the questionnaire. The second pre-test consisted of applying the questionnaire to a group of undergraduates to identify potential sources of error (poorly expressed items, aspects hard to understand, the order of the items, etc.). This allowed the definitive version of the questionnaire to be drafted. Table 1 presents the principal technical characteristics of the sample.

Table 2 lists the demographic and socioeconomic characteristics of the sample. One observes that approximately 49% were men and 51% women, ages between 18 and 25 years. Most (85%) were exclusively studying (not part-time), and most (78%) were part of households of 3–4 persons, in towns of more than 10 000 inhabitants, with net monthly household incomes between € 1500 and € 2500.

The questionnaire consisted of two distinct parts. The first comprised items relating to the variables described above that potentially influence the purchase intention of the remanufactured products under study. The second consisted of items relating to the respondent's demographic and socio-economic characteristics.

Table 1
Technical specifications of the sample.

Universe	80205 undergraduates
Geographical area	National (Spain)
Sample size	1529 surveys
Sample design	Convenience sample
Fieldwork	April–May 2011
Pre-tests	1st test: 13 experts (March 2011)
	2nd test: 56 undergraduates (March 2011)
Information analysis	Smart-PLS vn 2.0 and IBM SPSS Statistics vn 19.0

Table 2
Profile of the survey sample.

Demographics	Frequency ($n = 1529$)	Percentage
Gender		
Male	751	49.12%
Female	778	50.88%
Age		
Under age 21	792	51.80%
Age 21 to 25	631	41.27%
Age 26 to 30	73	4.77%
Age 30 or older	33	2.16%
Study Profile		
Arts and Humanities	11	0.72%
Social Sciences	1104	72.20%
Sciences	51	3.34%
Health Care Sciences	4	0.26%
Engineering	359	23.48%
Employment Status		
Student	1303	85.22%
Student and worker	226	14.78%
Household Size		
Between 1 and 2 persons	83	5.43%
Between 3 and 4 persons	1197	78.29%
Between 5 and 6 persons	234	15.30%
More than 6 persons	15	0.98%
Residence		
Rural area	282	14.84%
Urban area (≤10 000 inhabitants)	187	12.23%
Urban area (10 000–40 000 inhabitants)	470	30.74%
Urban area (≥40 000 inhabitants)	590	38.59%
Household Income		
Less than € 1500 per month	263	17.20%
Between € 1500 and 2500 per month	486	31.79%
Between € 2501 and 3500 per month	281	18.38%
Between € 3501 y 4500 per month	194	12.69%
More than € 4500 per month	152	9.94%
NA	153	10.01%

NA, not answered.

3. Results

For the analysis of the data, we used a structural equation model, applying partial least squares (PLS) optimization as is recommended in complex research problems in which there is little theoretical knowledge (Jöreskog and Wold, 1982; Wold, 1979), and in which both reflective and formative constructs are present (Cepeda and Roldán, 2005; Chin and Newsted, 1999; Lévy et al., 2006).

A PLS analysis consists of a two-stage procedure (Fornell et al., 1988; Lévy et al., 2006; Wold, 1966). First, the measurement instrument itself, the structural model, is evaluated for the reliability and validity of its scales. And second, the structural model is evaluated, analyzing the prior research hypotheses. According to Barclay et al. (1995): "*This sequence ensures that we have reliable and valid measures of constructs before attempting to draw conclusions regarding the relationships among the constructs*."

3.1. *Measurement model*

To check the reliability and validity of the measurement scales, we used the programs IBM SPSS Statistics vn 19 and Smart-PLS vn 2.0. One must note that it only makes sense to apply these tests to the reflective constructs which form part of the model (Chin, 1998). In our case, these are AP, SN, and PI. They are not applicable to formative constructs, which, in our case, are MMV and M. In this case, an analysis of the significance of their items' weights and a

Table 3
Measurement model: Reliability and Convergent Validity.

Items	Standardized loading/weight	*t*-value
Attitude towards Purchasing a remanufactured laptop (AP)		
Cronbach's alpha: 1.000; CRI: 1.000; AVE: 1.000		
(Likert scale: 1 = Very negative; 2 = Negative; 3 = Neutral; 4 = Positive; 5 = very positive)		
AP: Assesses the respondent's attitude if it were put to them that they were to purchase a remanufactured laptop.	1.000	0.00
Subjective Norm (SN)		
Cronbach's alpha: 0.6802; CRI: 0.8622; AVE: 0.7577		
(Likert scale: 1 = Absolutely disagree; 2 = Disagree; 3 = Indifferent; 4 = Agree; 5 = Absolutely Agree)		
SN1: My friends and people whose opinion I value would agree with my buying a remanufactured laptop due to its lower price.	0.873***	78.95
SN2: Most people who are important to me would think that I should not purchase a remanufactured laptop because I would be forgoing important advances (latest technology, performance, …)	0.868***	77.42
Motivations (M)		
Cronbach's alpha: N/A; CRI: N/A; AVE: N/A		
(Likert scale: 1 = Not important; 2 = Somehow important; 3 = Indifferent; 4 = Important; 5 = very important)		
M1: Degree of importance that the remanufactured laptop should incorporate the latest technology.	−0.498***	8.15
M2: Degree of importance that the remanufactured laptop is cheaper, even if this means giving up the most advanced technology.	0.518***	8.10
M3: Degree of importance that the remanufactured laptop should provide me with personal satisfaction in that it has less impact on the environment.	0.513***	8.18
Marketing Mix Variables (MMV)		
Cronbach's alpha: N/A; CRI: N/A; AVE: N/A		
(Likert scale: 1 = Not important; 2 = Somehow important; 3 = Indifferent; 4 = Important; 5 = very important)		
MMV1: Degree of importance that the remanufactured laptop should be available in conventional establishments.	−0.465**	2.49
MMV2: Degree of importance that the remanufactured laptop can be purchased through the Internet.	−0.279**	2.03
MMV3: Degree of importance of the design or appearance of the remanufactured laptop.	0.228**	2.00
MMV4: Degree of importance of the brand's reputation.	0.762***	3.29
Purchase Intention (PI)		
Cronbach's alpha: 0.5110; CRI: 0.7949; AVE: 0.6626		
(Likelihood of purchasing a remanufactured laptop: 1 = (0%–25%); 2 = [25%–50%); 3 = [50%–75%); 4 = [75%–100%))		
PI1: Intention to purchase a remanufactured laptop instead of an original laptop, taking into account the price and type of remanufacturer (original manufacturer or independent remanufacturer).	0.716***	26.35
PI2: Intention to purchase a remanufactured laptop instead of an original laptop.	0.902***	81.63

***$p < 0.01$; **$p < 0.05$; *$p < 0.10$; N/A: not applicable (formative constructs); CRI: Composite Reliability Index; AVE: Average Variance Extracted.

multicollinearity test were performed, as Henseler et al. (2009) recommend. All indicators show significant weights (Table 3) and the multicollinearity test confirms that there does not exist a high level of correlation between the constructs and their indicators, because the corresponding coefficient (Variance Inflation Factor) does not exceed the threshold value of 3.3 (Diamantopoulos and Siguaw, 2006).

For the reflective constructs, the reliability of the scales used to measure the variables of the proposed model was evaluated using the Composite Reliability Index (CRI) and Cronbach's alpha (α), as recommended by Fornell and Larcker (1981) and Wets et al. (1974). One observes in Table 3 that the CRI of all three reflective constructs exceeds the threshold value of 0.60 (Bagozzi and Yi, 1988). For the individual reliability of each reflective construct, measured by Cronbach's alpha, none exceeds the recommended value of 0.70 (Nunally and Bernstein, 1994). However, some authors argue that this criterion is too strict, and consider that values close to 0.50 may be acceptable in the early stages of model construction (Cepeda and Roldán, 2005). Also, the creator of the index (Cronbach and Shavelson, 2004) questions whether it is the best way to judge the reliability of a measuring instrument, and rather than other indices, such as the aforementioned CRI, might more adequately reflect the reliability of the measurement scales used. We therefore believe that the results obtained relative to the reliability of the measurement instrument are adequate.

The convergent validity of the measurement scales of the reflective constructs was analyzed by means of the Average Variance Extracted (AVE), and from the loadings of each indicator with respect to its own latent variable (outer loadings) and with respect to the other variables (cross loadings) (Table 3). All of the AVE values surpassed the recommended value of 0.50 (Fornell and Larcker, 1981), and the outer loadings exceeded the reference value of 0.60 (Bagozzi and Yi, 1988; Barclay et al., 1995; Chin, 1998).

The results of the cross loading analysis confirm that none of the indicators showed a significant loading (considering a loading to be significant when its value exceeds 0.70) in relation to a different latent variable to which these indicators were conceived (Table 4).

Finally, the discriminant validity was evaluated in terms of the average variance extracted (AVE), checking that the AVE for each factor or latent reflective construct (average variance shared between the construct and its indicators) was greater than its shared variance with any other factor or construct (the square of the estimated correlation between two constructs), as recommended by Fornell and Larcker (1981). The results (Table 5) showed the measurement model to satisfy the requirement of discriminant validity.

3.2. Structural model

After confirming the adequate fit of the measurement model, we evaluated the structural model (Fig. 2), and tested the research hypotheses using the Smart-PLS 2.0 software. The first step in evaluating the structural model was to compute the R^2 statistic, which indicates the amount of variance of the dependent variable explained by the predictor model variables. As one observes in Fig. 2, the value of R^2 for the model's dependent variable (PI) exceeds the reference value of 0.1 (Falk and Miller, 1992). The following step was to examine the structural model's predictive relevance by applying the statistical resampling technique known

Table 4
Measurement model: Convergent Validity (cross loadings).

	AP	SN	PI
AP	**1.0000**	0.2734	0.3513
SN1	0.2628	**0.8728**	0.2984
SN2	0.2128	**0.8681**	0.3231
PI1	0.2034	0.2039	**0.7159**
PI2	0.3466	0.3537	**0.9015**

Table 5
Measurement model: Discriminant Validity.

	AP	SN	M	MMV	PI
AP	**1.000**				
SN	0.273	**0.87**			
M	0.222	0.379	**N/A**		
MMV	−0.191	−0.234	−0.313	**N/A**	
PI	0.351	0.357	0.363	−0.261	**0.81**

N/A: not applicable (formative constructs). Off-diagonal elements: estimated correlations between factors. Diagonal elements: square root of the average variance extracted (AVE).

as blindfolding, using the Q^2 statistic (Geisser, 1975; Stone, 1974). In this procedure, part of the data is omitted when estimating a dependent latent variable from other independent latent variables, and then these same data are predicted using the parameters estimated previously. The process is repeated until each datum has been omitted and estimated. In view of the results (Table 6), the model can be considered to have explanatory power for the variable PI since the value of Q^2 (1 − SSE/SSO) is greater than zero.

The loadings of the reflective construct indicators, the weights of the formative construct indicators, and the β coefficients are standardized. R^2 (PI) $= 0.253$, $p < 0.01$.

Finally, we evaluated the significance of the factor loading relationships by the resampling technique known as bootstrapping (random sampling with replacement), as recommended by Chin (1998). In particular, 500 sub-samples were randomly generated from the original sample with replacement using a two-tailed Student's t distribution and 499 degrees of freedom ($n - 1$, where n is the number of sub-samples), with the following values:

Table 6
Structural model — explanatory power (blindfolding procedure).

Total	SSO	SSE	1 − SSE/SSO
PI	3058.0000	2559.5353	0.1630

SSE: sum of squares of the prediction errors; SSO: sum of squares of the observations.

$$t_{(0.10;499)} = 1.64791345;\quad t_{(0.05;499)} = 1.964726835;\quad t_{(0.01;499)} = 2.585711627.$$

We then proceeded to consider the respective tests of the research hypotheses (Table 7). One observes that all the coefficients were significant, so that the four hypotheses can be accepted.

4. Discussion

We would highlight from the results the proposed model's explanatory power on the variable under study, *Purchase Intention* (PI), together with the validity of the instrument of measurement. One can therefore state that both the model and the results adequately represent the intention to purchase this type of remanufactured product. In describing the profile of potential consumers of these products, we shall take as foundation the results for each of the variables that influence their intention to purchase, *i.e.*, attitude, subjective norm, motivations, and marketing mix variables (Fig. 3).

In this regard, it has been observed that a favorable attitude towards these products positively influences the purchase intention, so that one can assume that there is a segment of consumers who are favorably inclined to buying remanufactured computers, and that promotional actions of firms in the sector clearly should be directed at them. As Atasu et al. (2008) observed: "*The existence of a green consumer segment represents an important marketing opportunity for remanufacturers*". The consumer's principal social referents (family, friends, etc.) were shown to be key elements in the process of purchasing these products, at least with respect to the intention to purchase. Perhaps, therefore, firms in the sector should consider marketing activities designed to engage these referents in the purchasing process ("Bring a Friend" campaigns and the like) so

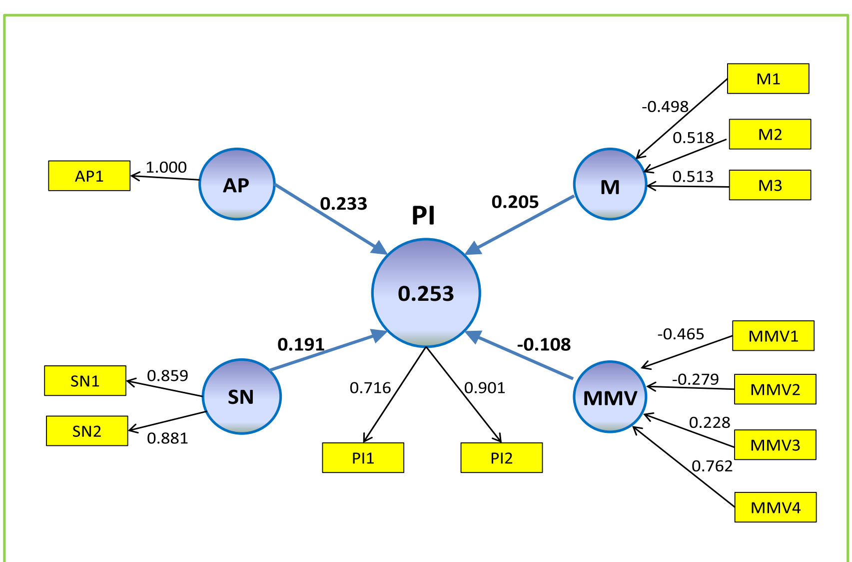

Fig. 2. Structural model.

Table 7
Structural model: Hypothesis tests.

Hypotheses	Standardized β	t-value (bootstrap)
H1: Attitude towards Purchasing (AP) → Purchase Intention (PI)	0.233***	8.92
H2: Subjective Norm (SN) → Purchase Intention (PI)	0.191***	7.98
H3: Motivations (M) → Purchase Intention (PI)	0.205***	8.41
H4: Marketing Mix Variables (MMV) → Purchase Intention (PI)	−0.108***	2.63

R^2 (PI) = 0.253; ***$p < 0.01$.

as to increase these consumers' likelihood of making a purchase. The study has also shown that motivations play a significant positive role in this purchase intention.

Overall, the respondents showed that price and environmental issues constitute positive motivations for their intention to purchase a remanufactured laptop, as has also been noted in other studies (among others, Agrawal et al., 2012; Atasu et al., 2010; Guide and Van Wassenhove, 2001). The technological aspects of these products seem to have a negative impact on the consumers' motivation. We might attribute this to the idea that consumers of this type of remanufactured product are not looking for a computer with the latest technology or with the greatest number of additional options. If they were, they would probably opt for purchasing an original product. Instead they are looking for a machine with a "good enough" performance at an attractive price. Thus, both price and the positive environmental aspects of remanufactured products should be important elements in firms' marketing policies so as to positively motivate consumers in their intention to purchase these products.

As regards the marketing mix variables, the results indeed pointed to the importance of this set of variables in describing the consumer's intention to purchase a remanufactured laptop computer. Nevertheless, there stood out the negative sign of the corresponding coefficient (Table 7). It is therefore interesting to consider the separate elements conforming this set of variables, *i.e.*, its four items MMV1-4 (Table 3). Thus, MMV1 and MMV2, both of which are related to distribution, have a negative sign, while MMV3 and MMV4 (design and brand reputation) have a positive sign. The two distribution items describe the importance the respondents give to the remanufactured product being marketed through conventional distribution channels and the Internet, thereby sharing the point of sale with the original products. We believe that, for their process of deciding to purchase a remanufactured laptop computer, the respondents have evaluated negatively the fact of not having specific distribution channels for these products in the sense that if the two types of product, original and remanufactured, share the same distribution channel, and are therefore available at the same point of sale, the consumer may well lean towards buying an original model rather than a remanufactured one.

5. Conclusions

With this work, our intention has been to contribute to the development of Closed-Loop Supply Chain research from a marketing perspective – specifically, from the point of view of consumer behavior. As mentioned in the Introduction, the objective was to identify the main characteristics describing the consumer profile for this type of remanufactured product, and thus offer firms in the sector information that is relevant to their business planning processes. In particular, it should be of use to them in establishing policies to develop and strengthen these particular markets. The results allow us to suggest some common characteristics of potential consumers of remanufactured computers. In particular, they have a favorable attitude towards this type of product, a clear motivation of respect for the environment, and a positive consideration of the opinion of their close social environment (family, friends, etc.) when making a purchase. In addition, they also regard both the product's price and the reputation of the OEM or of the remanufacturer as of particular importance for their intention to purchase these products. All these variables (price, brand reputation, motivation, attitude, social environment) can help us to explain the self-declared intention of purchasing a remanufactured laptop, and in this way to describe basic characteristics of the profile of the potential customers that can be used by companies in the development of marketing strategies for these markets.

Fig. 3. Key drivers in the behavior of potential consumers of remanufactured laptops.

We believe therefore that OEMs and remanufacturers could orient their marketing policies towards actions aimed at identifying consumers with a more favorable attitude towards these products. Their marketing campaigns could be directed not only at these consumers themselves but also at their closest social circles, which we have found to be an important referent in the intention to purchase. Other actions would be on the variables identified as "key drivers" such as the price and the OEM's or remanufacturer's reputation. Regarding product distribution, an analysis about whether the proper way of remanufactured product commercialization through specific retailers is necessary, because our results show that traditional retailers and the Internet do not meet the expectations of the remanufactured product consumer.

Naturally, we must acknowledge certain limitations of the present work. These are principally (i) the usage of a convenience sample makes it more difficult to use the results to infer broader conclusions that would be applicable to other products, markets for the same product (B2C vs. B2B), consumers or to firms in other sectors, and (ii) the participation of undergraduate students in the survey. With respect to the first question, this was clearly a matter of which we were aware from the start, but which should nevertheless not obscure the relevance of the results of the work. In spite of that, the hypotheses formulated and the results obtained are consistent with the results provided by similar works for other products (single-use cameras, power tools, toner cartridges, cell phones, or tires), as we mentioned in the text, so this approach could be extended to other products, customers and markets. Regarding the participation of students, we recognize that this is a controversial issue which has already been addressed in other studies, receiving arguments both pro (Rodney, 2011) and contra (Bello et al., 2009). In the present case, we believe that, because of their evident knowledge of the generic product under study, *i.e.*, computers, undergraduates may be regarded as a satisfactory example of the consumer of this type of product.

Finally, we would like to note that this is a novel study in that, to the best of our knowledge, no similar works have been published, and that it addresses some of the "gaps" in research that have been identified in the recent supply chain literature. Of course, there are still many issues to be analyzed – for example, those relating to the potential commercial activities and marketing policies that firms could institute. These should be contrasted empirically by means of case studies or the corresponding market research. It nevertheless represents a further step in the process of "closing the loop" in the supply chain by integrating the discipline of Marketing with that of Supply Chain Management in what we trust will be a fruitful path to follow.

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